Summary in Graph

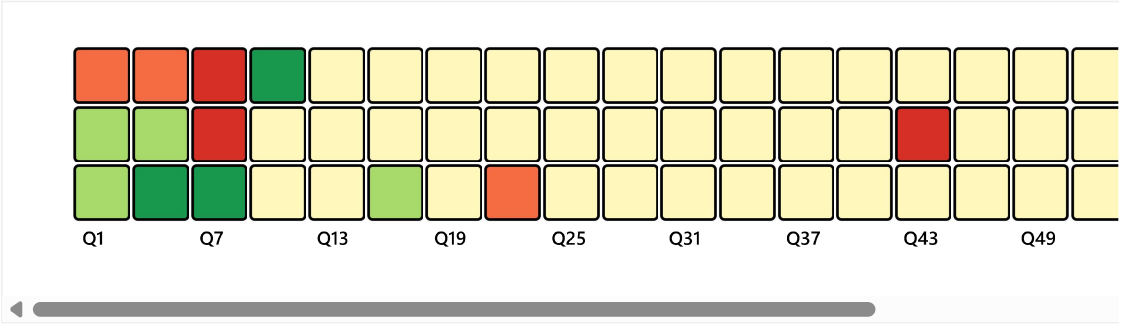
Exam Summary (GO Classes DA 2026 | All India Mock Test 1)

Qs. Attempted:	14 (7 + 7)	Correct Marks:	10 (4 + 6)
Correct Attempts:	7 (4 + 3)	Penalty Marks:	3 (1 + 2)
Incorrect Attempts:	7 (3 + 4)	Resultant Marks:	7 (3 + 4)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	180 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



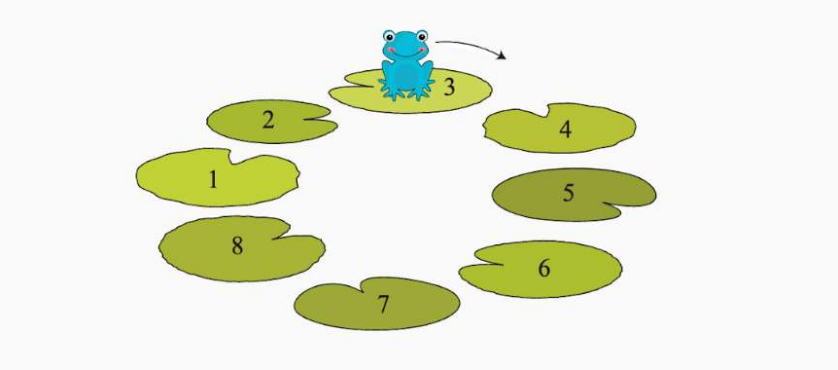
* Average Stats based on first-attempt stats of Top 10% (100) out of 1002 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Eight lily pads are arranged in a circle. A frog hops from one lily pad to the next in a clockwise direction. The frog begins on lily pad 3, as shown.



After 75 hops, which lily pad will the frog be on?

A. 5

B. 6

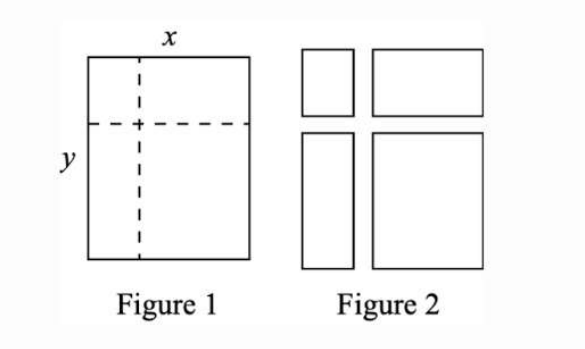
C. 7

D. 8

Your Answer: C | Correct Answer: B | Incorrect | Time taken: 01min 01sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Spatial Aptitude

A rectangle has width x and length y, as shown in Figure 1. The rectangle is cut along the horizontal and vertical dotted lines in Figure 1 to produce four smaller rectangles as shown in Figure 2.



The sum of the perimeters of these four rectangles in Figure 2 is 24. The value of $x + y$ is

A. 6

B. 8

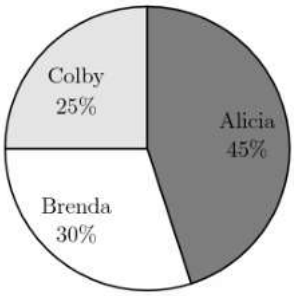
C. 9.6

D. 12

Your Answer: A | Correct Answer: A | Correct | Time taken: 03min 34sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Analytical Aptitude

Alicia, Brenda, and Colby were the candidates in a recent election for student president. The pie chart below shows how the votes were distributed among the three candidates. If Brenda received 36 votes, then how many votes were cast all together?



A. 84

B. 100

C. 106

D. 120

Your Answer: D | Correct Answer: D | Correct | Time taken: 00min 40sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Ara and Shea were once the same height. Since then Shea has grown 20% while Ara has grown half as many inches as Shea. Shea is now 60 inches tall. How tall, in inches, is Ara now?

A. 51

B. 52

C. 54

D. 55

Your Answer: A | Correct Answer: D | Incorrect | Time taken: 00min 26sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

The service at the restaurant was usually very attentive, but on this one occasion the waiter seemed to _____ the diners.

A. applaud

B. urge

C. ignore

D. restrain

Your Answer: C | Correct Answer: C | Correct | Time taken: 02min 23sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Along the route of a bicycle race, 7 water stations are evenly spaced between the start and finish lines, as shown in the figure below. There are also 2 repair stations evenly spaced between the start and finish lines. The 3rd water station is located 2 miles after the 1st repair station. How long is the race in miles?



A. 8

B. 16

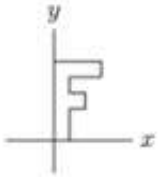
C. 24

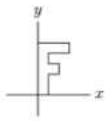
D. 48

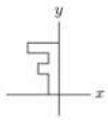
Your Answer: D | Correct Answer: D | Correct | Time taken: 12min 49sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Spatial Aptitude

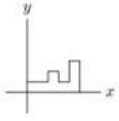
The letter F shown below is rotated 90° clockwise around the origin, then reflected in the y-axis, and then rotated a half turn around the origin. What is the final image?



A. 

B. 

C. 

D. 

Your Answer: C | Correct Answer: D | Incorrect | Time taken: 02min 58sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

Ms. Carroll promised that anyone who got all the multiple choice questions right on the upcoming exam would receive an A on the exam. Which one of these statements necessarily follows logically?

A. If Lewis did not receive an A , then he got all of the multiple choice questions wrong.

B. If Lewis did not receive an A, then he got at least one of the multiple choice questions wrong.

C. If Lewis got at least one of the multiple choice questions wrong, then he did not receive an A.

D. If Lewis received an A, then he got all of the multiple choice questions right.

Your Answer: C | Correct Answer: B | Incorrect | Time taken: 08min 32sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

In a recent basketball game, Shenille attempted only three-point shots and two-point shots. She was successful on 20% of her three-point shots and 30% of her two-point shots. Shenille attempted 30 shots. How many points did she score?

A. 12

B. 18

C. 24

D. 30

Your Answer: B | Correct Answer: B | Correct | Time taken: 03min 31sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

After seeing shocking films of animals maimed and tortured by traps of hunters, Marie concluded that purchasing a new fur coat would be _____.

A. mandatory

B. subliminal

C. glamorous

D. immoral

Your Answer: D | Correct Answer: D | Correct | Time taken: 00min 13sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following statements:

$$\text{Statement I} \quad : \quad \text{If } f(n) = O(n^2) \text{ and } g(n) = O(n^2), \text{ then } f(n) = O(g(n)).$$
$$\text{Statement II} \quad : \quad \text{If } f(n) = O(g(n)) \text{ and } g(n) = O(n^2), \text{ then } f(n) = O(n^2).$$

Which of the following options is correct?

A. Both Statement I and Statement II are true

B. Statement I is true and Statement II is false

C. Statement I is false and Statement II is true

D. Both Statement I and Statement II are false

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Suppose we run Dijkstra's algorithm on a graph with n vertices and $O(n \log n)$ edges. Assume the graph is represented using an adjacency list representation. What is the asymptotic running time of Dijkstra's algorithm in this case, if we use a binary heap for the priority queue? Express your answer as a function of n using $O(\cdot)$ notation.

A. $O(n \log n)$

B. $O((\log n)^2)$

C. $O(n(\log n)^2)$

D. $O(n^2)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Suppose you are given a list of n elements such that the location of each element is at most $\log(\log n)$ positions away from the location it would occupy in the fully sorted order.
Which of the following is the tightest asymptotic upper bound (using Big-Oh notation) on the running time of an algorithm that sorts this list?

A. $O(n)$

B. $O(n \log(\log n))$

C. $O(n \log n)$

D. $O\left((\log n)^2\right)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

A queue initially contains some elements.
In each step, one of the following operations is performed independently at random:

- With probability p : dequeue() - remove one element from the front.
- With probability $1-p$: enqueue() - add one new element at the rear.

What is the expected change in the number of elements in the queue after one operation?

A. $p-(1-2p)$

B. $1-2p$

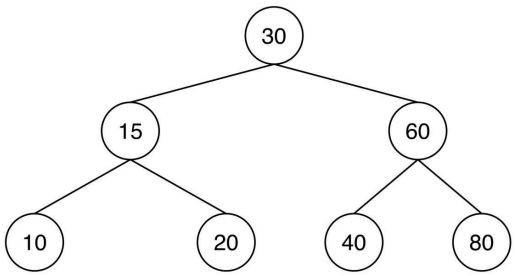
C. $2p-1$

D. $1-p$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

The following figure shows a binary search tree (BST) obtained by inserting distinct keys one by one into an initially empty tree.



Select the alternative that correctly represents an **insertion sequence** which generates *exactly the given BST*. Assume that all insertions are performed using the standard BST insertion rule.

A. 30, 15, 40, 10, 20, 60, 80

B. 30, 15, 40, 10, 20, 80, 60

C. 30, 15, 60, 10, 20, 40, 80

D. 30, 60, 20, 80, 15, 10, 40

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Consider the following binary classification dataset:

x_1	x_2	x_3	t
0	0	0	1
0	1	0	0
0	1	1	1
1	1	1	0

A linear classifier is defined as

$$f(\mathbf{x}) = \text{sign}(w_1x_1 + w_2x_2 + w_3x_3 + b)$$

where $t = 1$ corresponds to $f(\mathbf{x}) > 0$ and $t = 0$ corresponds to $f(\mathbf{x}) < 0$.

The classifier correctly classifies all training samples, and no sample lies on the decision boundary.

Which of the following inequalities must necessarily be satisfied by the weights and bias? (Select **all** that apply.)

A. $b > 0$
B. $w_2 + b < 0$
C. $w_2 + w_3 + b > 0$
D. $w_3 + b < 0$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider the K-fold cross validation on a linear regression model with a sufficiently large amount of training data. When K is large, the computational complexity of the K-fold cross validation with respect to K is of order

A. $\mathcal{O}(1/K)$.

B. $\mathcal{O}(K)$.

C. $\mathcal{O}(1)$.

D. $\mathcal{O}(K(K-1))$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Which of the following transformations to the data matrix $\mathbf{X}$ will affect the principal components obtained through PCA?

A. Adding a constant value to all elements of $\mathbf{X}$.

B. Adding an extra feature that is constant across all data points.

C. Multiplying one of the features of $\mathbf{X}$ by a constant.

D. None of the other options.

Your Answer: C | Correct Answer: C | Correct | Time taken: 06min 28sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & -3 \end{pmatrix}$$

and let I denote the 3×3 identity matrix.

If $6A^{-1} = aA^2 + bA + cI$, for real numbers $a, b, c \in \mathbb{R}$, then the ordered triple (a, b, c) is equal to:

A. $(1,\ 2,\ 1)$

B. $(4,\ 1,\ 1)$

C. $(1,\ -1,\ 2)$

D. $(1,\ 4,\ 1)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let A be an $m \times n$ real matrix such that AA^T is invertible.
Which of the following statements is/are necessarily true?

A. $\text{rank}(A) = m$

B. $\text{rank}(A) = n$

C. $m = n$

D. A is invertible

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider a real matrix A of size $m \times n$.
Which of the following statements is/are necessarily true?

A. If the row space of A is $\mathbb{R}^n$, then the column space of A is $\mathbb{R}^n$.

B. If the null space of A is $\mathbb{R}^n$, then the column space of A is the zero vector space $\{0\}$.

C. If the left null space of A is $\mathbb{R}^m$, then the column space of A is the zero vector space $\{0\}$.

D. If the row space of A is $\mathbb{R}^n$, then the column space of A is $\mathbb{R}^m$.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider a matrix A such that its null space $N(A)$ is spanned by a nonzero vector v.
Which of the following matrices represents the orthogonal projection onto the row space of A ?

A. $\frac{vv^T}{v^Tv}$

B. $I - \frac{vv^T}{v^Tv}$

C. $\frac{v^T v}{vv^T}$

D. $I + \frac{vv^T}{v^T v}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Consider a population of 50 units numbered $\{1, 2, \ldots, 50\}$.

The only possible samples are the following:

$$\{1\},\ \{1,2\},\ \{1,2,3\},\ \ldots,\ \{1,2,\ldots,50\}$$

One of these samples is selected at random, and each sample has the same probability of being chosen.

For a unit i, let p_i denote the probability that unit i is included in the selected sample.

For example, if $i = 8$, then p_8 is the probability that the selected sample contains the number 8.

Which of the following statements is correct?

A. $p_i = \dfrac{i}{50}$

B. $p_i = \dfrac{50 - i}{50}$

C. $p_i = \dfrac{1}{i}$

D. None of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X be a random variable which is symmetric about 0. Let F be the cumulative distribution function of X. Which of the following statements is always true?

A. $F(x) + F(-x) = 1$ for all $x \in \mathbb{R}$.

B. $F(x) - F(-x) = 0$ for all $\mathrm{x} \in \mathbb{R}$.

C. $F(x) + F(-x) = 1 + P(\mathrm{X} = x)$ for all $x \in \mathbb{R}$.

D. $F(x) + F(-x) = 1 - P(X = -x)$ for all $x \in \mathbb{R}$.

Your Answer: B | Correct Answer: C | Incorrect | Time taken: 01min 43sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let Y_1, Y_2 be independent random variables taking values -1 and $+1$ with probability $1/2$ each.

Define a sequence $\{X_n\}$ by

$$X_1 = Y_1, \quad X_2 = Y_2, \quad X_n = X_{n-1}X_{n-2} \text{ for } n \geq 3.$$

Find the value of $P(X_4 = 1)$.

A. $1/4$

B. $1/2$

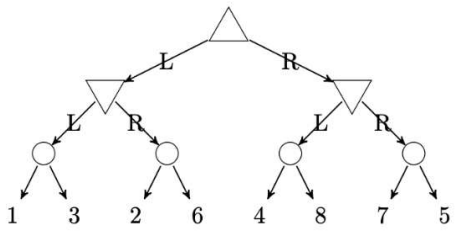
C. $3/4$

D. 1

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Numerical Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Consider a game with eight cards ($c \in \{1, 2, 3, 4, 5, 6, 7, 8\}$), sorted onto the table in four stacks of two cards each. MAX and MIN each know the contents of each stack, but they don't know which card is on top. The game proceeds as follows. First, MAX chooses either the left or the right pair of stacks. Second, MIN chooses either the left or the right stack, within the pair that MAX chose. Finally, the top card is revealed. MAX receives the face value of the card (c), and MIN receives $9 - c$. The resulting expectiminimax tree is as follows:

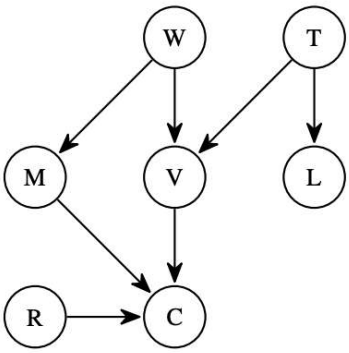


Assume that the two cards in each stack are equally likely. What is the value of the top MAX node?

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Pacman is going out to play a Pong tournament, and he wants to model all the uncertainty in the game in order to ensure victory. Consider the Bayes' net below. All random variables are binary.



Pacman is trying to compute $P(\mathrm{C} \mid +\mathrm{w}, -\mathrm{t})$. Pacman uses the standard variable elimination algorithm to eliminate L and M. Select the factor(s) needed to eliminate L and M.

A. $P(\mathrm{M} \mid +\mathrm{w})$

B. $P(\mathrm{C} \mid \mathrm{R}, +\mathrm{w}, \mathrm{V})$

C. $P(\mathrm{L} \mid -\mathrm{t})$

D. $P(\mathrm{C} \mid \mathrm{R}, \mathrm{M}, \mathrm{V})$

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Numerical Type | Award: 1 | Penalty: 0 | Programming in Python

Consider the following Python code:

```
        x0 = [0, None]
        x1 = [1, None]
        x2 = [2, x0]
        x3 = [3, x2]
5.      x3[1] = x3[1][1] = x1
        print(x2[1][0])
```

What will be the output of the above program?

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let f be a function such that $\lim_{h \to 0} \frac{f(2+h)-f(2)}{h} = 5$. Which of the following must be true?

I. f is continuous at $x = 2$
II. f is differentiable at $x = 2$
III. The derivative of f is continuous at $x = 2$

A. I only

B. II only

C. I and II only

D. I and III only

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let f be the function given by $f(x) = \ln(3 - x)$. The third-degree Taylor polynomial for f about $x = 2$ is

A. $-(x-2) + \frac{(x-2)^2}{2} - \frac{(x-2)^3}{3}$

B. $-(x-2) - \frac{(x-2)^2}{2} - \frac{(x-2)^3}{3}$

C. $(x-2) + (x-2)^2 + (x-2)^3$

D. $(x-2) + \frac{(x-2)^2}{2} + \frac{(x-2)^3}{3}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider a database relation R with attributes $\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}$ and F. What is the maximum number of candidate keys that R could have simultaneously?

Your Answer: | Correct Answer: 20 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Suppose we have a relation $R(A, B)$ with one tuple, both of whose components are NULL:

A	B
NULL	NULL

Then the result of:

```
SELECT A, COUNT(B)
FROM R
GROUP BY A;
```

A. $(\mathrm{NULL}, \mathrm{NULL})$

B. $(\mathrm{NULL}, 0)$

C. $(\mathrm{NULL}, 1)$

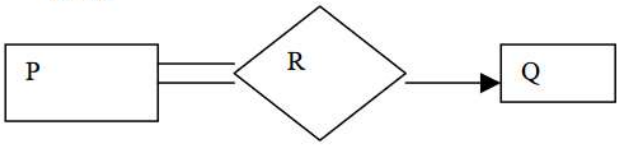
D. Empty Table

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

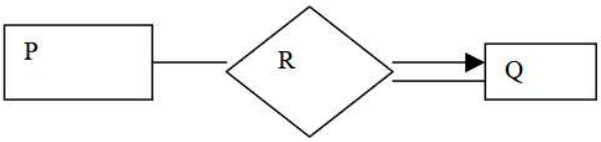
Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider the following two ER diagrams, $\mathrm{ER1}$ and $\mathrm{ER2}$:

ER1



ER2



The statement that the number of entities in entity set P must be greater than or equal to the number of entities in entity set Q holds for

A. $\mathrm{ER1}$ but not $\mathrm{ER2}$

B. $\mathrm{ER2}$ but not $\mathrm{ER1}$

C. Both $\mathrm{ER1}$ and $\mathrm{ER2}$

D. Neither $\mathrm{ER1}$ nor $\mathrm{ER2}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Consider the following proposition:

$$P : \forall x[Bx \rightarrow [Lx \wedge Cx]]$$

Mark the option that contains a proposition equivalent to $\neg\mathrm{P}$.

A. $\forall x \neg[Bx \rightarrow [Lx \wedge Cx]]$

B. $\exists x[Bx \wedge [\neg Lx \vee \neg Cx]]$

C. $\forall x[Bx \rightarrow \neg[Lx \wedge Cx]]$

D. $\exists x[\neg Bx \wedge [\neg Lx \vee \neg Cx]]$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

On which of the following intervals is function $f(x) = x^3 + x^2 - x$ both decreasing and concave up?

A. $\left(-\frac{1}{3}, \frac{1}{3}\right)$

B. $\left(-1, -\frac{1}{3}\right)$

C. $\left(\frac{1}{3}, \infty\right)$

D. $(-\infty, -1)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the following statements related to Depth-First Search (DFS):

Statement I:

Suppose DFS is run on an *undirected graph* with $|V|$ vertices. If a vertex v is discovered with $\text{start_time}(v) = 1$ and $\text{finish_time}(v) = 2|V|$, then the graph must be connected.

Statement II:

Suppose DFS is run on a *directed graph* with $|V|$ vertices. If a vertex v is discovered with $\text{start_time}(v) = 1$ and $\text{finish_time}(v) = 2|V|$, then every vertex is reachable from every other vertex.

Which of the following options is correct?

A. Both Statement I and Statement II are true
B. Statement I is true and Statement II is false
C. Statement I is false and Statement II is true
D. Both Statement I and Statement II are false

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Given two *sorted* arrays $A[1..m]$ and $B[1..n]$, we want to find the kth smallest element in the concatenation of the two arrays. Assume $k < m$ and $k < n$.

Consider the following recursive algorithm:

$$\begin{aligned}
&\text{Find_kth_elem}(A[1..m], B[1..n], k):\\
&\quad x \leftarrow A[k/2],\ y \leftarrow B[k/2]\\
&\quad \text{if } (x = y):\\
&\quad\quad \text{return } x\\
&\quad \text{if } (\underline{\quad}[P]\underline{\quad}):\\
&\quad\quad \text{return Find_kth_elem}\big(A[1..k/2],\ B[k/2+1..k],\ k/2\big)\\
&\quad \text{if } (\underline{\quad}[Q]\underline{\quad}):\\
&\quad\quad \text{return Find_kth_elem}\big(A[k/2+1..k],\ B[1..k/2],\ k/2\big)
\end{aligned}$$

Which of the following options is/are correct?

A. [P] = $x > y$, [Q] = $x < y$
B. [P] = $x < y$, [Q] = $x > y$

C. The worst-case time complexity of the algorithm is $O(\log m + \log n)$.

D. The worst-case time complexity of the algorithm is $O(\log(m + n))$.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Numerical Type | Award: 2 | Penalty: 0 | Data Structures

Consider a binary tree whose preorder and inorder traversals are both A, B, C, D, E.

The height of any rooted tree is defined as the maximum number of edges in the path from the root node to any leaf node.

What is the height of the tree?

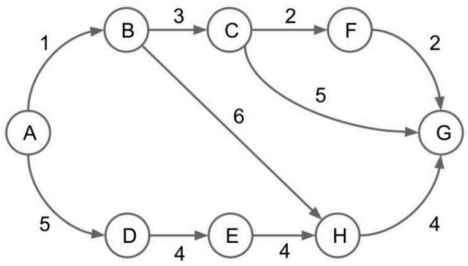
Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Recall that A^* is just Dijkstra's algorithm, except that vertices are given a priority equal to the sum of their Dijkstra priority (distance from the source) plus the heuristic distance, and also that we quit when the target is visited.

For the graph below, Suppose we use the following heuristic:

$$h(A) = 2$$
$$h(B) = 2$$
$$h(C) = 20$$
$$h(D) = 2$$
$$h(E) = 6$$
$$h(F) = 2$$
$$h(G) = 0$$
$$h(H) = 2$$



Which of the following paths is returned by $\mathbf{A}^*$ from $\mathbf{A}$ to $\mathbf{G}$?

A. $A \to B \to C \to F \to G$

B. $A \to B \to H \to G$

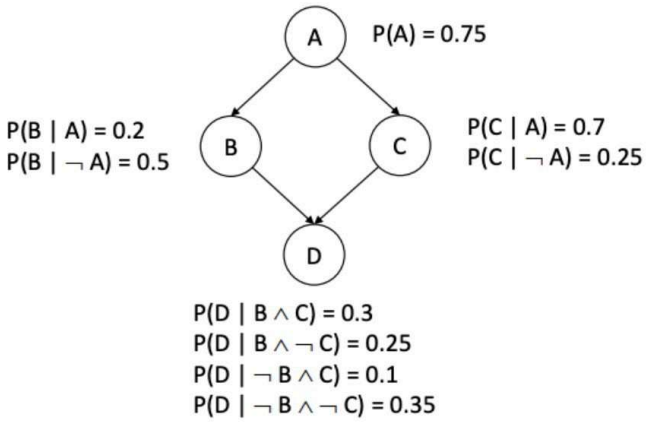
C. $A \to D \to E \to H \to G$

D. $A \to B \to C \to G$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Numerical Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Consider the following Bayesian network. A, B, C, and D are Boolean random variables. If we know that A is true, what is the probability of D being true (upto 3 decimal places)?



Your Answer: | Correct Answer: 0.197 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Consider an initially empty hash table with 32 buckets. The hash table does not resize in any of the scenarios below. Each subpart is independent, i.e., the hash table starts empty for each case. Do not assume anything about how the items are distributed across buckets.

After inserting items into the hash table, select all correct statements.

A. After inserting 16 items, the minimum possible number of items in bucket 0 is 0.

B. After inserting 16 items, the maximum possible number of items in bucket 0 is 16.

C. After inserting 16 items, bucket 0 must contain at least 1 item.

D. If 4 strings are inserted and each bucket is implemented as an unbalanced binary search tree, the worst-case total number of string comparisons required for all insertions is 6.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

We solve a classification problem with the perceptron algorithm. Labels are from the set {−1, +1}. We use $\eta = \frac{1}{2}$ for the learning rate. In one iteration of the algorithm, when updating the weights based on the example x, the weight vector w is changed so that its length is halved, but the direction and orientation are preserved.

Now consider that, instead of the perceptron, we use linear regression trained on labels {−1, +1} to do classification.

What would be the loss for the same example x if we used linear regression with weights w?

A. $1 - y\|w\|^2$
B. $1 - \|x\|^2$

C. $(\|w\|^2 + 1)^2$
D. $(y - \|x\|^2)^2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Consider the following labeled dataset in a two-dimensional input space:

$$\mathcal{D} = \{(\mathbf{x}^{(i)}, y^{(i)})\} = \{((1,0), +1),\ ((2,-3), -1),\ ((2,5), -1)\}.$$

This dataset is used as the training set for a perceptron. The input vectors are mapped to a six-dimensional feature space using the mapping

$$\phi(\mathbf{x}) = (1, x_1, x_2, x_1x_2, x_1^2, x_2^2).$$

The initial weight vector of the perceptron is given by

$$\mathbf{w} = (1, 0, 1, 2, -3, 0).$$

Using these initial weights, the perceptron makes predictions using the sign of $\mathbf{w} \cdot \phi(\mathbf{x})$. The loss incurred on a training example $(\mathbf{x}^{(i)}, y^{(i)})$ is defined as

$$\max\big(0,\ -y^{(i)}(\mathbf{w} \cdot \phi(\mathbf{x}^{(i)}))\big).$$

What is the total initial training loss of the perceptron on the training set before the start of training (i.e., with the given initial weights)?

Your Answer: | Correct Answer: 16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Our task is to classify whether an animal is a dog (class 0) or a cat (class 1) based on the following features:

- x_1 : height
- x_2 : length of whiskers
- x_3 : thickness of fur

We perform standard normal scaling on the training features so that they have a mean of zero and standard deviation of 1. We have trained a Logistic Regression model to determine the probability that the animal is a cat, $p(1 \mid \mathbf{x}, \mathbf{w})$.

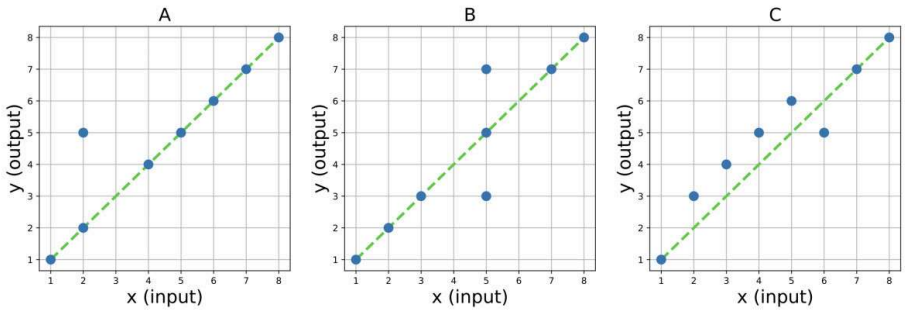
Our classifier learns that cats have a lower height and longer whiskers than dogs, while the thickness of fur is not relevant to the classification outcome. Which of the following is true about the weights $\mathbf{w}$ learned by the classifier?

A. $w_1 < w_2 < w_3$
B. $w_2 < w_3 < w_1$
C. $w_1 < w_3 < w_2$
D. $w_3 < w_1 < w_2$

Your Answer: B | Correct Answer: C | Incorrect | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

The following figure shows three different datasets (A, B, C) and the same linear model (dashed line) used to predict each of these datasets.



The dashed line represents a linear model mapping from $\mathbb{R}^1$ to $\mathbb{R}^1$. For each dataset, the Mean Squared Error (MSE) is computed between the predicted and true output values.

Choose the correct ordering with respect to the Mean Squared Error (MSE) loss.

A. $\text{MSE}_C > \text{MSE}_A > \text{MSE}_B$
B. $\text{MSE}_B > \text{MSE}_C > \text{MSE}_A$
C. $\text{MSE}_A > \text{MSE}_B > \text{MSE}_C$
D. At least two of the MSE losses are equal.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the following problem.

Let A be a block matrix with real entries, with each block B_{ij} of size 2×2, given by

$$A = \begin{bmatrix} B_{11} & B_{12} & B_{13} & B_{14} \\ 0 & B_{22} & B_{23} & B_{24} \\ 0 & 0 & B_{33} & B_{34} \\ 0 & 0 & 0 & B_{44} \end{bmatrix},$$

where

$$B_{11} = \begin{bmatrix} 2022 & 2021 \\ 2021 & 2022 \end{bmatrix}.$$

Which of the following statements is correct?

A. $\lambda = 1$ is an eigenvalue of A
B. $\lambda = 1$ is not an eigenvalue of A
C. $\lambda = 0$ is the only eigenvalue of A
D. All eigenvalues of A are strictly greater than 1

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

We are using Gradient Descent to minimize a **one–dimensional** loss function $\mathcal{L}(w)$, where $w \in \mathbb{R}$.

The loss function is *strictly convex*, and therefore it has a unique global minimum denoted by $w^\star$.

At iteration t, the current parameter value is w^t. Assume that $w^t > w^\star$.

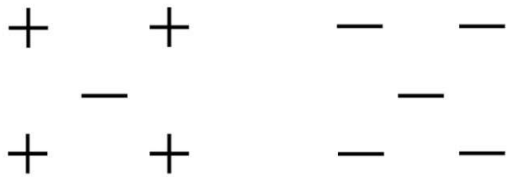
Which of the following statements correctly describes, in this **one–dimensional** setting, the sign of the gradient $\nabla \mathcal{L}(w^t)$ and the direction of the next Gradient Descent update w^{t+1}?

A. $\nabla \mathcal{L}(w^t) > 0$ and $w^{t+1} < w^t$
B. $\nabla \mathcal{L}(w^t) < 0$ and $w^{t+1} < w^t$
C. $\nabla \mathcal{L}(w^t) > 0$ and $w^{t+1} > w^t$
D. $\nabla \mathcal{L}(w^t) < 0$ and $w^{t+1} > w^t$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

We are given the following dataset as illustrated, and assume that "$+$" and "$-$" denotes the label of each data point:



The Leave-One-Out cross validation errors using 1-Nearest-Neighbor and 3-Nearest-Neighbor classifiers are:

A. $0, 5/10$.

B. $5/10, 1/10$.

C. $0, 1/10$.

D. $1/10, 5/10$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Given a matrix $\mathbf{X}$ of shape $D \times N$ with a singular value decomposition (SVD), $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$, suppose $\mathbf{X}$ has rank K and $\mathbf{A} = \mathbf{X}\mathbf{X}^\top$

Which one of the following statements is/are true?

A. The eigenvalues of $\mathbf{A}$ are the singular values of $\mathbf{X}$

B. $\mathbf{A}$ is positive semi-definite, i.e all eigenvalues of $\mathbf{A}$ are non-negative

C. The eigendecomposition of $\mathbf{A}$ is also a singular value decomposition (SVD) of $\mathbf{A}$

D. A vector $\mathbf{x}$ that can be expressed as a linear combination of the last $D - K$ columns of $\mathbf{U}$, i.e $\mathbf{x} = \sum_{i=K+1}^{D} \mathbf{w}_i \mathbf{u}_i$ (where $\mathbf{u}_i$ is the i-th column of $\mathbf{U}$), lies in the null space of $\mathbf{X}^\top$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let $u_1, u_2, u_3, v_1, v_2, v_3$ be vectors in $\mathbb{R}^4$. Let U be the span of $\{u_1, u_2, u_3\}$ and let V be the span of $\{v_1, v_2, v_3\}$.

Consider the following statements:

I. If the dimension of $U \cap V$ is 2 and the dimension of U is 3, then $\{v_1, v_2, v_3\}$ is linearly dependent.

II. If $U + V = \{u + v : u \in U, v \in V\} = \mathbb{R}^4$, then either $\{u_1, u_2, u_3\}$ is linearly independent or $\{v_1, v_2, v_3\}$ is linearly independent.

Which of the above statements is/are true?

A. Only I

B. Only II

C. Both I and II

D. Neither I nor II

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let A and B be real, invertible matrices such that

$$AB = -BA$$

Then which of the following statements is correct?

(Here $\mathrm{Tr}(\cdot)$ denotes the trace of a matrix.)

A. $\mathrm{Tr}(A) = \mathrm{Tr}(B) = 0$
B. $\mathrm{Tr}(A) = \mathrm{Tr}(B) = 1$
C. $\mathrm{Tr}(A) = 0, \mathrm{Tr}(B) = 1$
D. $\mathrm{Tr}(A) = 1, \mathrm{Tr}(B) = 0$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let A be a *real symmetric matrix* whose eigenvalues are $\lambda_1, \lambda_2, \ldots, \lambda_n$.

Define a polynomial by

$$p(t) = 1 + t.$$

For any vector x, let $\|x\|$ denote its **Euclidean length**.

Determine the value of

$$\max_{\|x\|=1} \| p(A)x \|.$$

A. $\max\{1 + \lambda_1,\ 1 + \lambda_2,\ \ldots,\ 1 + \lambda_n\}$

B. $\max\{|1 + \lambda_1|,\ |1 + \lambda_2|,\ \ldots,\ |1 + \lambda_n|\}$

C. $\min\{1+\lambda_1,\ 1+\lambda_2,\ \ldots,\ 1+\lambda_n\}$

D. $1+\lambda_1+\lambda_2+\cdots+\lambda_n$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let A be an $m \times n$ real matrix and consider the system $Ax = b$.

Which of the following statements are **correct**?

A. Any solution x of $Ax = b$ (if it exists) can be expressed as the sum of a vector from the **null space of** A and a vector from the **row space of** A.

B. The system $Ax = b$ is solvable if and only if b is orthogonal to every vector in the **null space of** A.

C. If A is a 4×3 matrix and the solutions of $Ax = b$ are not unique whenever they exist, then the rank of A must be exactly 2.

D. If x_1 and x_2 are two solutions of $Ax = b$, then $x_1 - x_2$ lies in the **null space of** A.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Consider the quadratic equation $x^2 + 2Ux + V = 0$, where U and V are chosen independently and at random from the set $\{1, 2, 3\}$, with each value having equal probability.

What is the probability that the given quadratic equation has both roots real?

A. $2/3$

B. $1/2$

C. $7/9$

D. $1/3$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose X_1, X_2, X_3 and X_4 are independent and identically distributed continuous random variables.

Consider the condition $X_4 > \max(X_1, X_2) > X_3$.

This means:

- X_4 is the largest among X_1, X_2, X_3, X_4
- X_3 is the smallest
- X_1 and X_2 lie in between (in any order)

Find $P(X_4 > X_3 > \max(X_1, X_2))$.

A. $1/6$

B. $1/8$

C. $1/12$

D. $1/24$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

A fair coin is tossed repeatedly.

Let X denote the number of Tails observed before the first Head occurs. Then

$$P(X = k) = P(\underbrace{T, T, \ldots, T}_{k \text{ times}}, H)$$

Let Y denote the number of Tails observed between the occurrence of the first Head and the occurrence of the second Head.

Which of the following statements about the distributions of X and Y are true?

A. $P(X = k) = 2^{-(k+1)}$ for $k = 0, 1, 2, \ldots$, and 0 otherwise.

B. $P(Y = k) = 2^{-(k+1)}$ for $k = 0, 1, 2, \ldots$, and 0 otherwise.

C. $P(X = k) = 2^{-k}$ for $k = 1, 2, 3, \ldots$, and 0 otherwise.

D. $P(Y = k) = (k+1)2^{-(k+2)}$ for $k = 0, 1, 2, \ldots$, and 0 otherwise.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Numerical Type | Award: 2 | Penalty: 0 | Probability

Let $X = (X_1, X_2, X_3)^T$ be a random vector with distribution $N_3(\mu, \Sigma)$, that is, a *three-dimensional multivariate normal* distribution with mean vector μ and covariance matrix Σ.

The mean vector is given by

$$\mu = (1, 1, 1),$$

and the covariance matrix is

$$\Sigma = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & c \\ 1 & c & 2 \end{pmatrix}.$$

Suppose that the random variables X_2 and $-X_1 + X_2 - X_3$ are independent.

Then the value of c is _____.

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Consider performing tree search on a search graph using Iterative Deepening Depth-First Search (IDDFS). Let:

- $\text{depth}(n)$ be the depth of search node n,
- $\text{cost}(n)$ be the total path cost from the start node to n,
- G_d be a goal node with minimum depth,
- G_c be a goal node with minimum total path cost.

Recall that iterative deepening repeatedly performs depth-limited DFS, increasing the depth limit by 1 each iteration, and terminates as soon as a goal node is found.

Which of the following conditions are guaranteed to be true for every node n that could be expanded during the search?

A. $\text{cost}(n) \leq \text{cost}(G_c)$

B. $\text{cost}(n) \leq \text{cost}(G_d)$

C. $\text{depth}(n) \leq \text{depth}(G_c)$

D. $\text{depth}(n) \leq \text{depth}(G_d)$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in Python

Consider the following two functions:

Code A

```
n = 5

def func():
    global n
    if n == 0:
        return
    n -= 1
    print(n, end=" ")
    func()

func()
```

Code B

```
def func(n=[5]):
    if n[0] == 0:
        return
    n[0] -= 1
    print(n[0], end=" ")
    func(n)

func()
```

What will be the outputs of Code A and Code B?

A. Code A: `4 3 2 1 0` Code B: `4 3 2 1 0`
B. Code A: `4 3 2 1 0` Code B: `5 4 3 2 1`
C. Code A: `3 2 1 0` Code B: `4 3 2 1 0`
D. Code A: `4 3 2 1 0` Code B: `4 4 4 4 4`

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in Python

Consider the following Python code:

```
def fun(x, y):
    if x == 0:
        return y
    p = x - 1
    q = y + 2
    x, y = p, q
    return fun(x, y)

print(fun(3, 1))
```

What will be the output of the above program?

A. 5

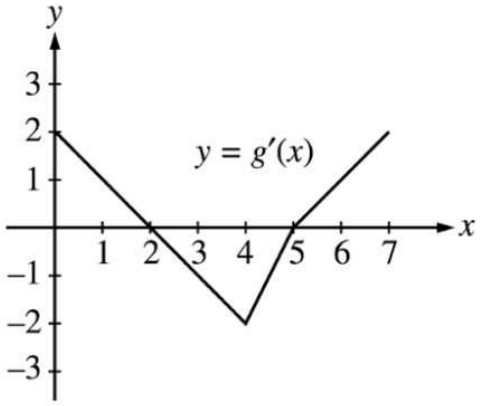
B. 6

C. 7

D. 8

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus



The graph of the derivative g' of a function g with domain $[0, 7]$ is shown above. Which of the following must be true about g ?

I. g has a local maximum value at $x = 0$ and a local minimum value at $x = 4$.

II. g has a local maximum value at $x = 2$ and a local minimum value at $x = 5$.

III. $g(2) = g(5)$.

A. I only

B. II only

C. III only

D. I and II

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Calculate the order p of the internal node of a B+ tree (order is same as fan-out). Suppose the search field is $\mathrm{V} = 8$ bytes long, the block size is $\mathrm{B} = 1024$ bytes, and a block pointer is $\mathrm{P} = 2$ bytes. An internal node of the B+ tree can have up to p tree pointers and $\mathrm{p} - 1$ search field values, and these must fit into a single block. What is p _____ ?

Your Answer: | Correct Answer: 103 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider the following relational database tables R, S and T :

R

A	B
1	2
3	2
5	6
7	8
9	8

S

B	C
6	2
2	4
8	1
8	3
2	5

T

A	C
7	1
1	2
9	3
5	4
3	5

What is the number of tuples in the resulting table of $\pi_{\mathrm{A,B}}(\mathbf{R} \bowtie \mathbf{T}) \bowtie \pi_{\mathrm{B,C}}(\mathbf{S} \bowtie \mathbf{T})$?

Your Answer: | Correct Answer: 9 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

If it is true that the following three statements are true,

$$p \Rightarrow q$$
$$r \Rightarrow s$$
$$(p \wedge t) \Leftrightarrow r$$

then it is true that:

A. $\neg \mathrm{s} \Rightarrow (t \vee p)$

B. $\neg \mathrm{r} \Rightarrow \neg \mathrm{s}$

C. $\neg q \Rightarrow \neg r$

D. All of the previous answers.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

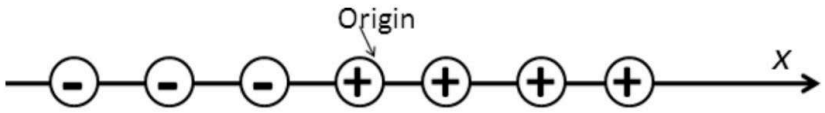
Consider the following soft-margin Support Vector Machine (SVM) optimization problem:

$$\min_{\{w,b\}} \frac{1}{2} w^T w + C \sum_{i=1}^{m} \varepsilon_i$$

subject to

$$y_i(w^T x_i + b) \geq 1 - \varepsilon_i, \qquad \varepsilon_i \geq 0 \ \ \forall i$$

A one-dimensional dataset is shown in the figure below, where points marked with "$+$" belong to the positive class and points marked with "$-$" belong to the negative class.



If $C = 0$, meaning that only the margin size is optimized and misclassification is not penalized, how many support vectors does the resulting SVM classifier have?

Your Answer: 2 | Correct Answer: 7 | Incorrect | Time taken: 09min 16sec | Discuss

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